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Urban traffic externalities: Quasi-experimental evidence from housing prices☆

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ABSTRACT

This paper exploits a quasi-experiment to value the benefits of reducing urban traffic externalities. As a source of exogenous variation we use the opening of a new bypass in The Hague, the Netherlands that reduced traffic on a number of local streets, leaving others unaffected. We make use of a unique longitudinal dataset on traffic densities and residential sales, and calculate the effect of the change in traffic nuisance on housing prices. We find that, on average, a reduction of 50% in traffic density induces a 1.4% increase in housing prices. Reductions in traffic nuisance are valued much more positively when the traffic density is already high. Furthermore, our results indicate that traffic nuisance effects are likely to be biased in cross-sectional studies.

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1. Introduction

Data for the European Union indicate that passenger kilometres driven by car rose by a factor of two and a half between 1970 and 2000 and have increased by yet another ten percent since (European Commission, 2002, 2012). Different cities have recently introduced or strengthened policies aimed at reducing the incoming automobile traffic. Although automobiles have become much cleaner and quieter in the recent decennia, they still are a leading source of greenhouse gases, local air pollution (Parry et al., 2007) and noise annoyance (CE, 2011).[1] In urban areas, the negative externalities caused by road traffic are especially harmful due to the high concentration of population. For example, before the introduction of the congestion charge in London in 2003, ninety percent of the city residents surveyed opined that there was too much traffic in London (Rocol, 2000). This paper exploits a quasi-experiment to produce new and improved evidence on the benefits of reducing urban traffic externalities. As a source of exogenous variation we use the opening in 2003 of a new highway in the west of the Netherlands that provided bypass capacity for through traffic heading to the third largest Dutch city, The Hague. The bypass considerably reduced traffic on a number of local streets in the suburbs of The Hague, leaving others unaffected. We exploit these induced traffic changes on local streets in a hedonic price model of residential sales to derive how much people value reductions in traffic nuisance.

With this paper we contribute to a growing literature on the effects of policies aimed at the reduction of urban traffic. The existing empirical papers mainly focus on the effects of recently introduced congestion charges and report considerable reductions in traffic volume in London (Leape, 2006) and Stockholm (Eliasson, 2009; Kopp and Prud'homme, 2010). We contribute to this literature by examining how urban citizens value the decrease in traffic externalities. A number of theoretical papers analyse the effects and relative performance of a broader set of policy measures. De Borger and Proost (2013) consider tolls, bypass capacity to guide traffic around the city centre and capacity reduction measures such as speed limits and road bumps. Karamychev and Van Reeven (2011) investigate park-and-ride facilities that aim to intercept motorists

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[1] In 2008 fifteen percent of the EU-27 population (some eighty million people) was exposed to potentially harmful road noise levels above sixty decibels, compared to one and a half percent of the population exposed to railway and airplane noise above sixty decibels.

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from travelling into the city. We complement this literature by providing empirical evidence on the possible benefits of these policies; this evidence can be used to support the optimal choice of the policy instrument and its design in practice.

Our paper is also related to the quasi-experimental research on the valuation of environmental goods (see Boes and Nüesch, 2011, and the references therein for airport noise; Palmquist, 1982, for highway noise; Chay and Greenstone, 2005, for air pollution; Davis, 2011, for the nuisance from power plants). Our paper is the first to combine a quasi-experiment and a fixed effect hedonic price estimation to study the valuation of local traffic nuisance. Palmquist studies traffic externalities using variation in urban noise levels induced by the construction of a highway through a town. He does not, however, take into account the environmental effects of changes in the local traffic flows brought about by the realization of the new highway. These latter effects constitute the focus of our paper. Furthermore, in our study, the measured effect of traffic nuisance accounts not only for noise, but for the whole array of various local externalities induced by local urban traffic. As described in Parry et al. (2007), these mutually correlated externalities include, among other things, congestion, noise, local pollution and accident risks. By focussing on the external effects of traffic, our paper is furthermore indirectly related to the small quasi-experimental literature on the accessibility effects of new transport infrastructure.[2]

Finally, our paper is connected to the literature that studies the anticipation effects of improvements in transport infrastructure, and their capitalization in the land prices. Existing literature presents no unambiguous conclusion. McDonald and Osuji (1995) and McMillen and McDonald (2004) find evidence that the future increase in rail accessibility was already partly capitalized in the land prices three years before the opening of the new Chicago transit line. Billings (2011) finds no support for an immediate announcement effect from the decision about light rail construction in Charlotte, NC; he argues that land markets need several years to adjust to the new information. Gibbons and Machin (2005) argue, however, that owner–occupiers discount the future benefits of transport improvements very heavily so that anticipation effects will, in most cases, be quite limited. All these papers focus on the anticipation effects connected to accessibility. Our paper sheds some light on the anticipation effects of other changes in location characteristics that new infrastructure can bring about, such as a relief of local traffic.

Our identification strategy exploits exogenous variation in traffic density on the local streets within a radius of some 5 km on both sides of the bypass. We use a unique dataset including (i) micro data on local traffic densities on the streets in these suburbs, and (ii) individual residential sales data on affected and unaffected streets before and after the opening of the bypass. A noteworthy feature of our dataset is the continuous and longitudinal data on traffic density. Affected streets not only differ in the level of traffic density in the before period, but also in the magnitude of the induced density change. This variation allows us to progress beyond a simple comparison of housing prices on affected and unaffected streets and estimate a functional relationship between traffic density and housing prices. Exact information on the location of the dwellings makes it furthermore possible to include fixed effects on a very low level of aggregation (street segment) and to identify the effect of traffic nuisance from the before-after variation in traffic nuisance and housing prices. Finally, our data covers a large enough time period before the opening of the bypass to be able to shed some light on the anticipation effects.

Taking advantage of our detailed data on traffic density, we detect a statistically significant positive effect of a decrease in traffic density on the value of houses within 20 m from the affected streets. The estimated elasticity of housing prices to traffic density is on average −0.02 for houses adjacent to the street. It is factor 1.5 smaller for dwellings located further away. For houses on very busy roads the effect of traffic density turns out to be much stronger: For streets with traffic flows above 15,000 cars per day an elasticity coefficient of −0.09 is estimated. The baseline estimate is robust to various tests of the model we perform.

Furthermore, we do not find any significant anticipation effects up to three years prior to the opening of the bypass. This finding is in line with the anecdotic evidence. Consultations with local transportation experts suggest that residents had trouble realizing ex ante that the bypass would influence traffic flows on local streets. What is more, the construction of the bypass had, for many decades, been opposed by local residents, who argued, among other things, that it would not reduce through traffic in the surroundings (Hobma, 2000).

Finally we investigate how much the use of the quasi-experimental methodology confers advantages in terms of pinning down the valuation of traffic nuisance in comparison with more conventional methods based on pooled cross-sectional regressions. The results of conventional studies are likely to suffer from the omitted variables bias, as unobserved neighbourhood characteristics tend to be correlated with both the housing prices and the environmental good (e.g. Greenstone and Gayer, 2009; Boes and Nüesch, 2011). A quasi-experimental study like ours has the advantage of avoiding this problem. We use our data to estimate pooled cross-sectional models and find considerably lower effects of traffic nuisance (about 4 times lower), suggesting a positive correlation between traffic density and unobserved neighbourhood amenities. This result may have implications for the valuation of road traffic externalities in transport policy appraisals in different countries. Currently, these valuations largely build on the outcomes of the conventional hedonic studies on the effects of traffic externalities (see Odgaard et al., 2005; Navrud, 2004).

The paper proceeds as follows: In the next section we describe the institutional framework of the events leading to the local traffic density changes and explain our identification strategy. Section 3 deals with the data used, and Section 4 presents the results of the estimations. Section 5 discusses the policy implications of the results for the valuation of traffic nuisance. Section 6 concludes.

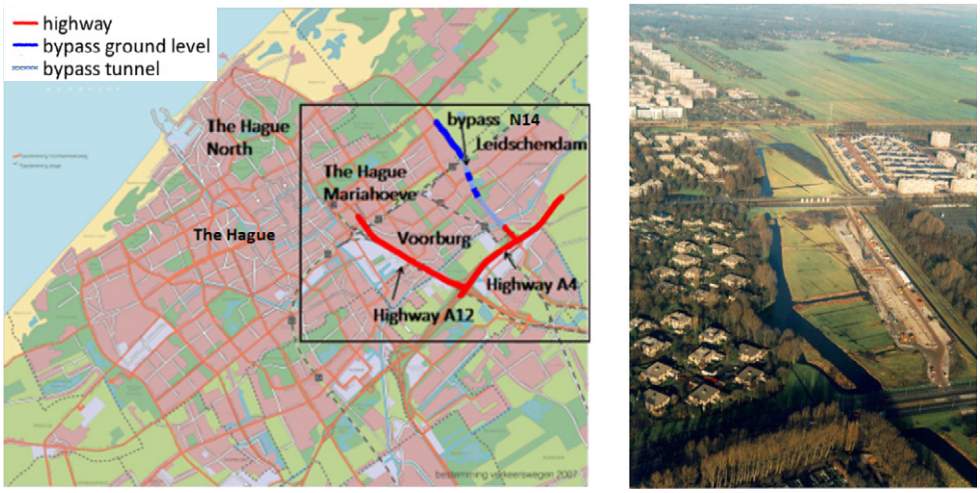
2. Changes in local traffic and identification strategy

2.1. Planning process of the bypass

In November 2003 a new highway N14 was opened in the Netherlands, providing an additional access road to the third largest Dutch city, The Hague. The Hague is located near the sea (see Fig. 1 left panel), making a beltway impossible. The only existing access road, highway A12, had been congested for a long time inducing through traffic to The Hague North to use the local streets of the suburbs Leidschendam, Voorburg and The Hague Mariahoeve. The new highway offered bypass capacity relieving traffic on both the access road A12, and on local streets in the suburbs.

The lead time to the opening of the bypass was extremely long. It first appeared in the plans of the Ministry of Transportation in 1938 as a means to improve highway connection to the northern part of The Hague. The precise location of the bypass was determined in 1956, and at that time a land reservation for it was made at its current location, between the urban neighbourhoods of Leidschendam and Voorburg and along the urban neighbourhood of The Hague Mariahoeve. However, due to successful local opposition, construction was deferred for decades and the land reservation remained open space (see Fig. 1 right panel). The local city councils argued that it was unclear that the new bypass would alleviate traffic problems in The Hague. In 1974 a small majority in the Federal Parliament voted against financing the bypass.

[2] Klaiber and Smith (2010) study accessibility benefits of a new highway. Gibbons and Machin (2005) and Billings (2011) study these benefits for new train stations. Hurst and West (2014) examine the effects of a new light rail transit on land use in the surroundings.



Source: Ministry of Transportation and Environment

Fig. 1. The research region (left) and the land reservation for the bypass in 1999.

In the beginning of 1980s the Ministry of Transportation made a new attempt to force construction. As the municipalities of the suburbs kept opposing the construction, a study was performed on alternative solutions to traffic problems in The Hague region including small extensions to the existing roads and investments in public transport. The bypass emerged as a preferred solution. Possible alternative alignments for the bypass also got some attention, but they appeared much less effective in solving traffic problems in the northern part of The Hague and were soon rejected.

By 1994 the city councils of The Hague and Leidschendam supported the bypass, while Voorburg remained in opposition. By the 1990s the residential neighbourhoods of Voorburg had grown very close to the land reservation. Furthermore, the land reserved for the bypass was owned by the municipality of Voorburg and, by that time, had become an attractive location for residential construction. Under federal pressure to accept the bypass, in 1996 Voorburg suggested a solution in which the highway would run in tunnels between Voorburg and Leidschendam.[3] The tunnels would be financed by the residential construction on the top of the tunnels of some 700 houses and a park. In August 1998 the plan was accepted and a decision was made to start the construction works in January 1999.

The bypass opened in January 2003 and absorbed a considerable part of the through traffic from the local streets of Leidschendam, Voorburg and The Hague Mariahoeve. We will exploit these changes in local traffic to estimate the valuation people attach to traffic nuisance. An important concern in many quasi-experimental studies of transport infrastructure is that the location of a transportation investment is endogenous to the traffic problems of the neighbourhood where it takes place. In our case, however, there are strong arguments why the location of the bypass can be seen as exogenous to traffic density at the neighbourhood level in Leidschendam, Voorburg and Mariahoeve at the time of announcement. First, the decision about the location of the bypass was mainly based on its effectiveness as a means of improving traffic accessibility of The Hague North. The bypass was meant to relieve the only other existing highway connection to The Hague, highway A12. Traffic relief on local streets was a side-effect rather than the main goal of the bypass. Due to the location of The Hague near the sea and the impossibility of a beltway, there were not many alternative alignments for the bypass available. Second, it was in 1950s that the precise location of the bypass was determined; construction was, however, postponed for many decades due to local protests. In 1956 a land reservation for the bypass was made, that was eventually used for construction. The location of the bypass was thus not influenced by the specific neighbourhood traffic developments in the study area in the 1990s, the time the construction decision was taken.

2.2. Induced changes in local traffic density

The bypass runs in three tunnels through two urbanized suburbs of The Hague, Leidschendam and Voorburg, and has there two intersections with the urban road network (Fig. 2). Through the Hague neighbourhood Mariahoeve it runs above ground, the nuisance being mitigated by a sound wall. Although relatively small in geographical scale, with its length of 3.5 km, the bypass absorbed a large part of the through traffic that had previously used local streets in these suburbs. Traffic density on local through streets decreased on average by half (from 8.3 thousand to 4.2 thousand cars per day); some affected streets lost up to 90% of the before-traffic and up to 15 thousand cars per day. Fig. 3 (lower panel) illustrates the traffic relief experienced in different local through streets. This relief is major when compared with the general trend in the west of the Netherlands, where the traffic density remained more or less stable after 2003. See Table 1.

We will use the described variation in local traffic flows in the before and after period to estimate how the decrease in urban traffic nuisance is capitalized in housing prices. Fig. 3 (upper panel) illustrates the location of the housing stock relative to the through streets that experienced changes in traffic density. A noteworthy feature of our data is the continuous and longitudinal data on traffic density. Affected streets not only differ in the level of traffic density in the before period, but also in the magnitude of the induced density change. Housing prices on these affected streets can be expected to adjust in line with the respective density changes. Some 20% of the houses in our dataset (1715 out of 8740) are located on the affected streets. Houses on unaffected non-through streets inside the residential neighbourhoods can be seen as a control group. Given the size of our dataset, including houses on

[3] Once persuaded of the effectiveness of the bypass, the municipal council of The Hague showed no opposition to the bypass running above the ground along its neighbourhood Mariahoeve. In The Hague the expected disamenity from the ground-level bypass was limited. Residential neighbourhoods were located at some distance from the land reservation (some 100 m of open space separates houses from the bypass). Besides, the closest houses stood perpendicular to the planned highway, and, on the other side of the bypass, there was agricultural land only.

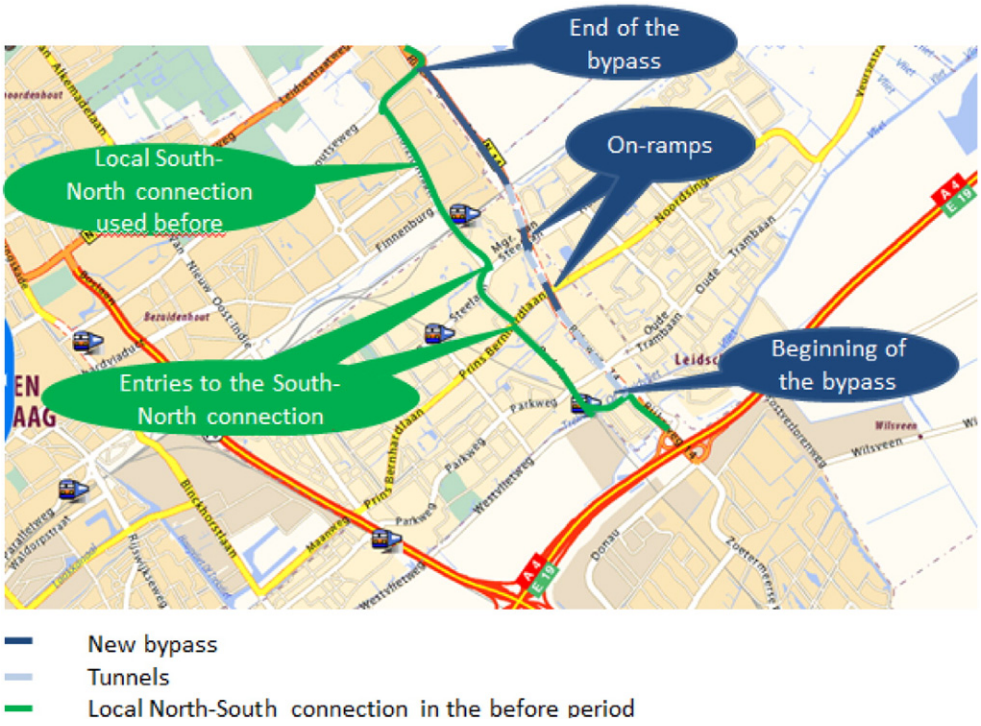


Fig. 2. New bypass and the South–North local connection used in the before-period.

unaffected streets in the analysis is desirable as it improves the statistical significance of the estimation results. We will show, however, that the point estimates do not hinge on the definition of the control group. The estimated price elasticity to traffic density proves to be similar for the whole estimation sample that includes the control group, and for the sample including only houses on the affected streets.

The next section describes the identification strategy in more detail. But first we address two general methodological concerns that might complicate the identification of the effects.

Next to changes in local traffic, other things have changed in much of the treated area due to the construction of the bypass. First, road access has improved: the bypass introduced a faster connection to the intercity road network and to The Hague. Second, the bypass has led to a change in neighbourhood amenities for houses in its direct vicinity. Indeed, a land reservation (open space, see Fig. 1 right panel), made space for a residential area in Leidschendam-Voorburg and a road with a sound wall in Mariahoeve. Thus it is very important to think about what treatment each particular house has got. If the variation in housing prices induced by either of two above mechanisms is highly correlated with the variation induced by traffic density changes, this would lead to biased estimates. However, the highly disaggregated data we have on traffic density changes allow us to easily control for the possibility of this bias. Variation in traffic density changes in our data is on the level of a street segment between two junctions (see Fig. 3 lower panel). Consultations with the transportation experts from local authorities[4] revealed that travel time improvements due to the faster new connection to The Hague do not vary much within larger neighbourhoods[5] and can thus be accounted for by the inclusion of neighbourhood-specific time trends. Fig. 2 gives the intuition for this. Before the bypass was built, the local street parallel to it was largely used as a connection to The Hague. The entries to this street from local roads are located close to the new on-ramps. It is thus likely that the routes chosen by the locals to get to The Hague did not change much due to the bypass. Thus, the absolute accessibility of the study area changed, but the relative accessibility of its different parts as compared to each other stayed practically unchanged. Accessibility gains (due to less overloaded local streets) are then similar within larger neighbourhoods. However, to be on the safe side, we will also control in the estimation for the distance to the on-ramps of the bypass.

We furthermore account for the neighbourhood amenity effect of the bypass by including specific time trends for the neighbourhoods located near the tunnelled and the ground part of it. We expect—if any—positive externalities near the new residential area on the top of the tunnel and negative externalities near the ground part of the bypass. The negative externalities are likely to be limited however as the ground level part of the bypass runs through agricultural land on some distance from the residential neighbourhood The Hague Mariahoeve. Some 100 m of open space separates houses from the bypass, and the first houses are located perpendicular to the bypass. The noise nuisance was furthermore mitigated by a sound wall.

The construction of the bypass started in January 1999, and, in November 2003, the first cars made use of the new connection. If the residents of surrounding towns anticipated the effects of reduced local traffic densities before these became effective, the prices could have already partly adjusted to the new situation by 2003. To get more insight into this issue, we perform a simple difference-in-difference estimation comparing the price appreciation trends for houses that experienced a traffic decrease of more than 50% after the opening of the bypass with houses that remained unaffected (see Fig. 4). If anticipation effect took place, then it should be seen most clearly in the streets with the largest improvements in traffic nuisance. The analysis suggests that before 2003, both groups of houses followed a parallel price appreciation trend. In the last quarter of 2003, simultaneously with the opening of the bypass, housing prices on streets with decreased traffic experienced a permanent upward jump. This preliminary analysis does not provide evidence of an anticipation effect, neither in 1998, when the decision

[4] We are especially grateful to Don de Greef (Municipality Leidschendam-Voorburg) and Louis Eggen (Municipality The Hague).

[5] The extra time savings of the residents of streets with decreased traffic (e.g. due to easier and closer parking possibilities) we attribute to the effects of decreased local traffic density.

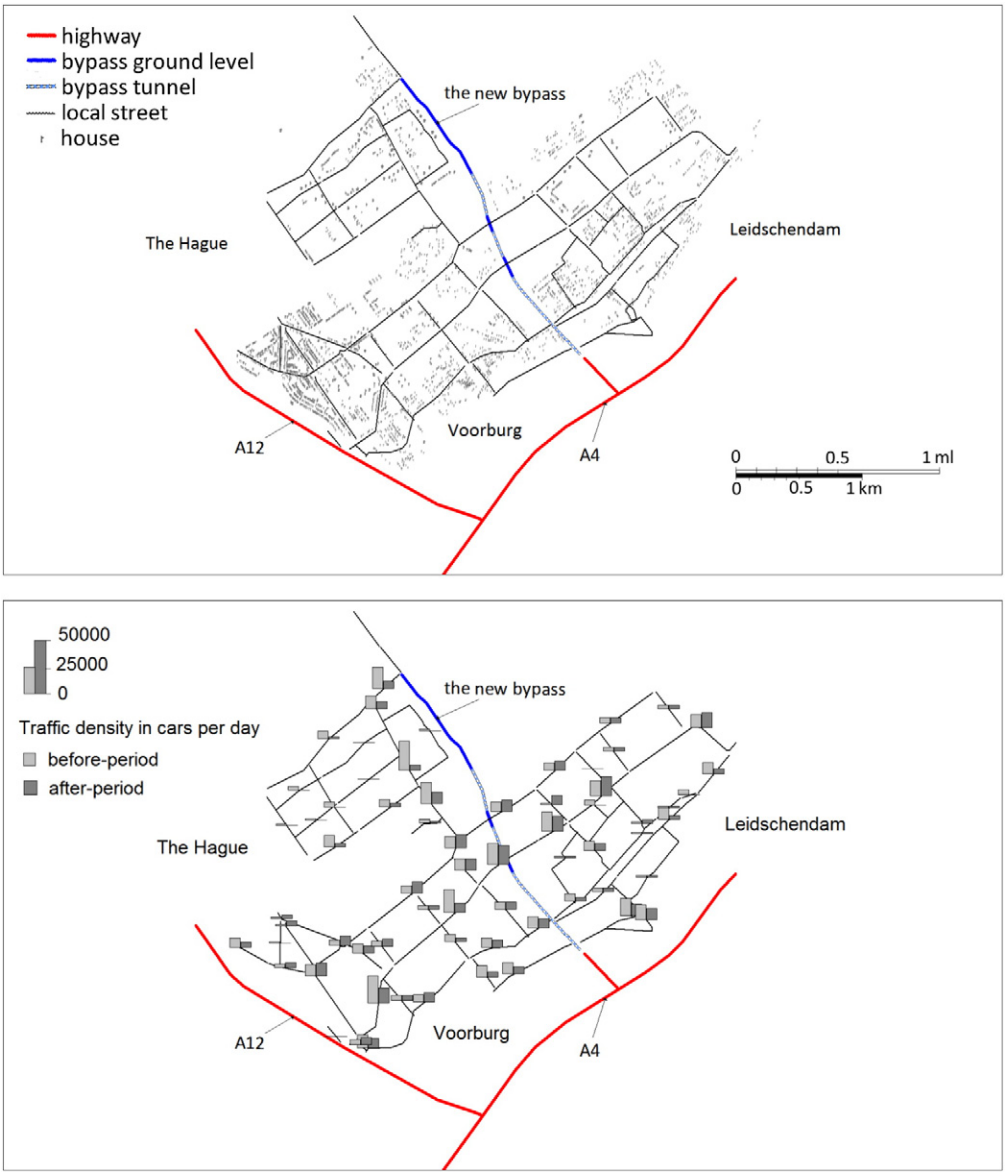


Fig. 3. Location of houses with respect to the affected through streets (upper panel). Changes in traffic density on through street segments (lower panel).

was announced, nor later. Consultations with local transportation authorities revealed that residents had trouble realizing ex ante that the bypass would influence traffic flows on local streets. What is more, the construction of the bypass had been for many decades opposed by the local residents, who argued, among other things, that it would not reduce through traffic in the surroundings (Hobma, 2000). We will however also formally test for the existence of the anticipation effect in Section 4.

2.3. Identification strategy

A starting point for our analysis is a location fixed effect Eq. (1) relating transaction prices of houses P to the traffic density D on the street where the house is located. To control for differences in other location characteristics of the houses sold, we include time-invariant fixed effects for the smallest administrative units in the Netherlands (zip code PC6 units) that include 10 to 15 dwellings located on the same side of a street.

Table 1
Traffic density in the West of the Netherlands 1998–2006.

	1998	1999	2000	2001	2002	2003	2004	2005	2006
Index traffic density	94	98	100	102	103	105	104	104	105

Source: Statistics Netherlands.

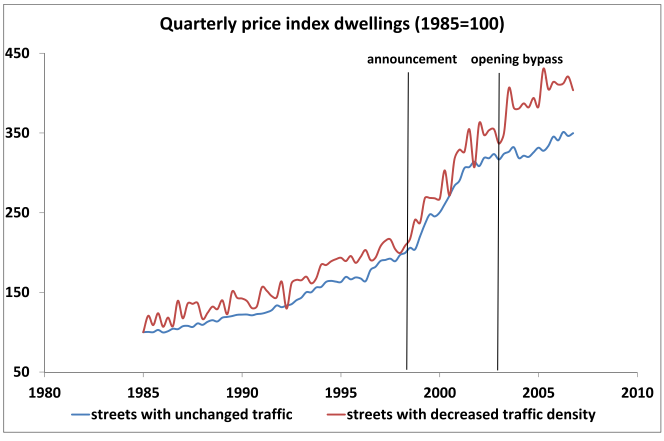


Fig. 4. Decreased traffic nuisance capitalized in housing prices after the opening of the bypass.

This ensures that the traffic density discount β is estimated from the variation within these units. We furthermore control for the variation in the structural characteristics of the houses sold by including covariates X for these structural characteristics. Finally, the general time trend and neighbourhood specific time trends are accounted for.

$$\ln P_{ijt} = \alpha + \beta \ln D_{jt} + f_j + \gamma_1 X_{it} + \gamma_2 Y_t + \gamma_3 I_{neighb} Y_t + \varepsilon_{ijt} \quad (1)$$

where P_{ijt} is the price of dwelling i located in zip code unit j in year t; D_{jt} is the traffic density in zip code unit j; f_j is the zip code fixed effect; X_{it} is a vector of the structural housing attributes of dwelling i in year t (such as: size of the dwelling, number of rooms, presence of a garage, etc.); Y_t are year dummies; $I_{neighb}Y_t$ are neighbourhood-specific time trends that account for possible differences between neighbourhoods in the accessibility effects and neighbourhood amenity effects of the bypass (see discussion above); β and γ are coefficients to be estimated and ε_{ijt} is the residual idiosyncratic term of an individual house.

After consultations with local transportation authorities we distinguished three larger neighbourhoods that can differ in accessibility effects[6] and two smaller neighbourhoods, adjacent to the bypass, that could have experienced (positive or negative) external effects from the bypass. Vector I_{neighb} thus contains five neighbourhood-specific dummies that take value one if the house is located in a specific neighbourhood and zero otherwise. As an alternative measure of accessibility improvement we include the distance to the on-ramps as an explanatory variable.

A double-log specification of the relationship between the housing price and the traffic density allows interpreting the coefficient by density as price elasticity. The choice to express density in logarithms is based on the following consideration: The level of noise, which is the major source of traffic nuisance, is commonly expressed by a logarithmic measure of the effective sound pressure relative to a reference value (measured in decibels).[7]

Our estimates will be based on the price differences between the periods 1998–2003 and 2004–2006. Before 1998 the data on structural characteristics of houses are limited. After 2006 other important changes in the transport network in the area took place.

We exploit a demand shift due to a decrease in local traffic to estimate the willingness to pay to avoid traffic nuisance. The extent to which this demand shift will be reflected in prices depends on the housing supply elasticity. If housing supply is perfectly inelastic, reductions in traffic will be fully capitalized into home prices. Table 2 shows that the housing supply in the study area changed marginally (by 2.5%) between the before and the after period. The main part of this increase was due to the new residential area built on the top of the bypass tunnel. In our estimations we account for the positive external effect these new properties could have had on the houses in the surroundings.[8] Appendix A shows furthermore that the composition of the housing supply did not change much between the before and after period.

Endogeneity of housing transactions is another concern. A higher turnover in the study area and a change in the composition of the population following the change in local traffic would make it hard to interpret results as willingness to pay. Table 2 reports some descriptives of the population in the study area suggesting there were no significant changes in population composition. Appendix A offers a more extensive discussion.

[6] These are: town of Voorburg, town of Leidschendam and The Hague neighbourhood Mariahoeve.

[7] As a robustness check, we have also run regression (1) with traffic density included in a linear way. The main insights from our study proved robust to this adjustment.

[8] As our identification strategy is based on the comparison of residential prices before and after the bypass, the prices of newly built properties will not enter the estimation and cannot bias the results.

Table 2
Population and housing supply in the study area.

	1998	2006
Houses (thousands)	42.5	43.5
% apartments	70%	70%
% rented property	n/a	52%
% families with children	27%	26%
% 25–65 years old	55%	56%
Average income per household (×1000)	14	15

Source: Statistics Netherlands.

2.4. Testing for anticipation effects

We allow anticipation to take place up to three years before the opening of the bypass. We amend Eq. (1) with a term: $\beta_2 I_{2001\text{-}2003} D_{j,after}/D_{j,before}$, where $D_{j,after}/D_{j,before}$ is the relative change in traffic density between the before and after period, β_2 is the coefficient to be estimated and $I_{2001-2003}$ is an indicator variable taking the value one if the house was sold in the years 2001–2003 and the value zero otherwise.

Eq. (1) becomes therefore:

$$\ln P_{ijt} = \alpha + \beta_1 \ln D_{jt} + \beta_2 I_{2001-2003} \frac{D_{j.after}}{D_{j.before}} + f_j + \gamma_1 X_{it} + \gamma_2 Y_t + \gamma_3 I_{neighb} Y_t + \varepsilon_{ijt}. \quad (2)$$

In Eq. (2), we allow the expected change in traffic nuisance in the after-period 2004–2006 to capitalize in the housing prices already in the years t = 2001, 2002 and 2003. We expect the coefficient β_2 to be negative as expected decrease in traffic nuisance should capitalize in higher housing prices.

3. Data

This research uses two main sources of data: (i) information on housing sales between 1998 and 2006 in the towns Voorburg, Leidschendam and the neighbourhood Mariahoeve in The Hague, and (ii) data on traffic densities on the through streets in the same region. Micro data on the houses sold were provided by the Dutch Association of Real Estate Brokers (NVM).[9] These data include the transaction date, the transaction price and extended information on housing attributes, such as age, construction descriptors (e.g. type of heating, presence of a built-in garage, etc.) and various dimensional attributes (such as the size of the living area, the number of rooms, etc.) We know the exact location of each house.[10] Our identification strategy (see Section 2) is based on repeated sales on the level of a zip code PC6 unit.[11] Consequently, we exclude the zip code units that are present in the before- or after-period only (in this way the new construction on the top of the tunnel is not affecting the estimates). This results in an unbalanced panel consisting of 8740 observations within 1005 different zip code units. Roughly 60% of the observations refer to the period before the bypass became operational (1998–2003), 40% of the observations refer to the after-period (2004–2006). In Section 4 we provide a robustness check with an alternative data source, the Netherlands' Cadastre and Public Register Agency (Kadaster). Kadaster registers all sales transactions

[9] Nationwide, around 75% of all residential property sales is performed through a real estate broker who is member of NVM. We focus on sales transactions and not rent transactions as in the Netherlands the rented sector is heavily regulated, ¾ of it being social housing. Price formation for social housing takes place in a non-market way that hardly accounts of the locational characteristics of the properties. The NVM-sales data we use are similar to those used in Van Ommeren et al. (2011).

[10] For their help with geocoding the data, we thank the department Spatial Economics of the VU University Amsterdam.

[11] PC6 is the smallest administrative unit containing 10-15 houses located on the same side of a street.

(as compared to some 75% registered by NVM), but it records very limited information on housing characteristics. The Kadaster data allow performing a repeated sales estimation on the level of an individual dwelling. The balanced panel used contains 1338 houses.

Detailed data on traffic densities of approximately 80 through street segments in the research area in the before- and after-period were made available by the Municipality of Leidschendam-Voorburg. Based on the knowledge of the exact geographical location of every dwelling and every street segment, the two datasets could be linked. Fig. 3 upper panel shows the geographical reach of our research area and the location of the dwellings in our dataset in relation to the through streets and the bypass.

Our dataset contains measurements of traffic density at two points in time: 1999 and 2006. To obtain figures for the years 1998 and 2000–2005 we use the data of Statistics Netherlands on the growth rate of traffic on larger streets in the western part of the Netherlands. Table 1 above reports these growth rates. As a robustness check in Section 4 we perform an estimation on 1999 and 2006 only.

As described in Section 2, we control for the possible neighbourhood amenity effects of the bypass on its direct surroundings. Due to mitigation measures (a tunnel and a sound wall), if these effects were present, they were very local. As Fig. 2 shows, a large part of the bypass was built in tunnels on the top of which new residential housing and a park were realized. Nuisance from the ground part of the bypass was mitigated by a sound wall. We allow dwellings within a distance of approximately 400 m on both sides of the bypass to experience a (positive or negative) change in spatial quality; the precise borders of the proximity area are determined manually in GIS using, as guidelines, the location of larger streets.[12] According to this definition, around 5% of the houses in our dataset lie in the area where neighbourhood amenity effects of the bypass might be present.

As a first step towards estimating the impact of the traffic density changes on housing prices, we compare prices and traffic densities for three groups of observations: (i) the observations on through streets that experienced a fall in traffic density after the opening of the bypass, (ii) the observations on through streets that experienced a rise in traffic density after the opening and (iii) the observations not lying on one of the affected streets.[13] For each group of observations, the before (1998–2003) and after (2004–2006) means are reported.

Table 3 suggests that the opening of the bypass led to considerable decreases (on average 4100 cars a day) and only modest increases (1300 cars a day on average) in traffic density. The related percentage price changes are in line with the expectations. Non-affected dwellings showed a price rise of 21% between the two periods. Dwellings with a rise in traffic density, experienced a smaller price appreciation, 19%, while the price of dwellings with a fall in the traffic density grew more, with some 26%. Appendix A reports the full descriptive statistics of the data.

Table 3
Housing price and traffic density in the before and after periods[a].

	Dwellings with a fall in traffic density		Dwellings with a rise in traffic density		Non-affected dwellings	
Variable	Before	After	Before	After	Before	After
Housing price (×1000 euro)	159	200	189	224	168	204
Traffic density (×1000 cars/day)	8.3	4.2	5.6	6.9	n/a	n/a
# observations	799	415	281	150	4409	2686

[a] The before period is reported in prices of 2000, the after period is reported in prices of 2004.

4. How does traffic density affect housing prices?

In this section we estimate model (1) and some extensions.[14] We pay especial attention to the heterogeneity of the effect of traffic density. Furthermore, various robustness checks are presented.

Model 1 of Table 4 below presents the baseline elasticity of home prices to the traffic density on the street where these dwellings are located. This elasticity equals −0.02. This implies that a 1% decrease in the traffic density results in a 0.02% increase in the prices of houses affected by this traffic. A 50% decrease in traffic density leads to a 1.4% increase in the housing prices. The coefficient is highly significant.

It is quite possible that dwellings located further away from affected streets also profited from reduction in traffic nuisance. We test this hypothesis and find that the effect of the traffic density change reaches as far as 20 m from the affected street. Model 2 of Table 4 reports the benefits for houses within 20 m from the street to be 1.5 times smaller than the benefits for houses adjacent to the street.

In the estimation, we account for heterogeneity of treatment effects on different houses. The opening of the bypass may have changed the amenity neighbourhood effects for houses in the vicinity. We correct for this by including neighbourhood-specific time trends. The point estimates in Table 4 suggest that dwellings near the tunnel and the residential construction on its top experienced a positive effect equal to 1.8% of the price, and dwellings near the ground part of the bypass lost some 0.9% of the price despite the sound wall. The signs of the estimates are in line with the expectations, however the coefficients are insignificantly different from zero.

Furthermore, the bypass improved road accessibility. We account for accessibility effects of the new bypass by including neighbourhood-specific time trends. The underlying assumption is that within a neighbourhood (defined as the towns Voorburg and Leidschendam and The Hague neighbourhood Mariahoeve) the accessibility change is similar. In model 3 of Table 4 we test this assumption by explicitly including the effect of the distance to the on-ramps of the bypass. In line with the expectations of the local transportation experts (see the discussion in Section 2 above) we do not find any significant effect.

To go deeper into the heterogeneity of treatment effects, we estimate the valuation of traffic nuisance reductions as a function of the distance to the bypass. Ideally we would like to distinguish between the distance to the on-ramps, the distance to the tunnel and the distance to the ground segments. These distances however have a correlation coefficient of 0.95–0.99. For this reason we focus on the cross-effect of density with the distance to the on-ramps (model 4 in Table 4). The results suggest no significant effect. This provides support to the additive way in which we model how various changes in the location characteristics affect the prices of treated houses.

In Appendix C we show that the estimated traffic nuisance effect is robust across space, for dwellings in different price segments and for different types of dwellings (apartments and one-family homes). Below another important source of heterogeneity is discussed, namely, the initial level of nuisance.

Noise is an important source of traffic nuisance. Literature on the effects of traffic noise suggests that, starting with a noise level of around

[12] We experimented with different definitions of the neighbourhood amenity effect (larger or smaller distances to the bypass), the results are robust to these definitions.

[13] Non-affected dwellings are houses located inside residential neighbourhoods, on non-through streets. For these streets no traffic density measurements were performed as this density is low and mainly influenced by local traffic flows. For the non-through streets we make two alternative assumptions: (i) traffic density did not change between the before- and after- period; (ii) traffic density grew with the average rate for the west of the Netherlands. Our results are robust to these assumptions.

[14] To account for possible correlation between the error terms within a zip code unit, we apply the clustered errors correction as suggested by Angrist and Pischke (2008).

Table 4
Estimations of the price elasticity to traffic density[a].

Standard errors are in parentheses					
	Baseline model (model 1)	Include further away located houses (model 2)	Include effect distance to on-ramps (model 3)	Cross-effect density with distance to on-ramp (model 4)	High and medium density streets separate, (model 5)
Elasticity to density:					
• houses adjacent to affected street, all densities	−0.019***	−0.021***	−0.019***	−0.019	
	(0.005)	(0.005)	(0.005)	(0.020)	
• houses adjacent to affected street, medium densities					−0.018***
					(0.005)
• houses adjacent to affected street, high densities					−0.090***
					(0.032)
• houses not-adjacent to affected street, within 20 m from it		−0.013**			
		(0.006)			
• houses not-adjacent to affected street, within 20–40 m from it		−0.005			
		(0.004)			
Effect of distance to the on-ramps (km)			0.005	0.005	
			(0.006)	(0.006)	
Cross-effect ln density with distance to the on-ramp (per km distance)				−0.001	
				(0.012)	
Neighbourhood amenity effect bypass, houses adjacent to tunnel	0.018	0.018	0.025	0.024	0.019
	(0.013)	(0.013)	(0.015)	(0.015)	(0.013)
Neighbourhood amenity effect bypass, houses adjacent to the ground part	−0.009	−0.005	−0.005	−0.005	−0.009
	(0.042)	(0.042)	(0.042)	(0.043)	(0.043)
Neighbourhood-specific time trend	Yes	Yes	Yes	Yes	Yes
Location (PC6) fixed effects	Yes	Yes	Yes	Yes	Yes
House fixed effects	No	No	No	No	No
Year dummies	Yes	Yes	Yes	Yes	Yes
Structural housing characteristics	Yes	Yes	Yes	Yes	Yes
R^2 within (adjusted)	0.76	0.76	0.76	0.76	0.76
# observations affected streets	1715	1715	1715	1715	1715
# observations unaffected streets	7025	7025	7025	7025	7025
# fixed effects	1005	1005	1005	1005	1005

[a] Coefficient estimates for the structural housing characteristics are reported in the Appendix B.
* 10% significance.
** 5% significance.
*** 1% significance.

45 decibels and higher, people experience noise as annoyance. Above a level of 60 decibels noise may lead not only to annoyance but also to health problems such as sleep disturbance or heart disease (World Health Organization, 2011). This insight suggests that the elasticity of the housing price to traffic density may be different for low/medium and high values of the density. To test this hypothesis we adjust Eq. (1) allowing for a piecewise relationship between the logarithm home price and the logarithm traffic density. We allow for a kink at the density level of 15,000 cars per day that corresponds to the noise level of approximately 60 decibels.[15]

Model 5 in Table 4 reports the results of this estimation. For the low and medium densities, the estimated value of price elasticity does not significantly change and stays highly significant. For the high values of traffic density, a factor 5 larger elasticity value of −0.09 is found. The result for the high densities is significant at 5% and in line with the intuition. Nevertheless, the point estimate for the high densities should be treated with some caution. In our dataset the data coverage for the high traffic densities is rather thin: only 15% of the 1725 affected houses faced in the before period a traffic density above 15,000 cars a day.

[15] This calculation has been made using the SMRI method, which is applied in the Netherlands to calculate the noise levels corresponding to different values of traffic density. Note that equal traffic density values can lead to different noise levels depending on the local circumstances (e.g. the type of asphalt). In our calculation the average values suggested at the site www.stillerverkeer.nl were used.

4.1. Robustness checks

A six digit postcode is a small spatial unit containing on average 10 to 15 houses. By including postcode unit fixed effects, we can thus be quite confident that possible omitted location variables have been accounted for. To perform a formal test, we do a repeated sales analysis. The data from the Association of Real Estate Brokers (NVM) used in the specifications of Table 4 inhibit a house from being consistently followed through time. For this reason, the repeated sales analysis is based on the sales transactions from the Netherlands' Cadastre and Public Register Agency (Kadaster). The Kadaster is a public agency that registers the actual selling price of all houses, it contains 100% of the housing sales (as compared to 75% in the data by NVM). The registered housing characteristics in the Kadaster data are however very limited. In our study area there are 1338 houses that have been sold, both in the before (1998–2003) and after (2004–2006) period. Some 20% of these transactions are located on affected streets. Model 1a in Table 5 reports the resulting estimates for the baseline model. The point estimate of the price elasticity on traffic density stays well within the confidence bounds of the baseline fixed effects estimate. Its significance is lower, however, due to the smaller sample size of the dataset. This robustness check suggests that postcode fixed effect estimation indeed accounts well for the omitted variable bias.

We would also like to establish whether houses differ on observables between treated and untreated streets in a way that is correlated with the treatment. For that purpose we estimate the model

Table 5
Robustness checks.

Standard errors are in parentheses						
	Baseline model 1	Repeated sales (model 1a)	No structural house characteristics (model 1b)	Affected streets only (model 1c)	Two years only, 1999 and 2006 (model 1d)	Anticipation effect (model 1e)
Elasticity to density	−0.019***	−0.014	−0.017**	−0.020**	−0.015	0.016
	(0.005)	(0.010)	(0.008)	(0.009)	(0.020)	(0.015)
Anticipation effect (per 10% density change)						−0.001
						(0.005)
Neighbourhood-specific time trend	Yes	Yes	Yes	Yes	Yes	Yes
Location (PC6) fixed effects	Yes	No	Yes	Yes	Yes	Yes
House fixed effects	No	Yes	No	No	No	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Structural housing characteristics	Yes	No	No	Yes	Yes	Yes
R^2 within (adjusted)	0.76	0.82	0.48	0.79	0.82	0.72
# observations affected streets	1715	534	1715	1715	215	1715
# observations unaffected streets	7025	2142	7025	0	810	7025
# fixed effects	1005	1338	1005	186	306	1005

* 10% significance.
** 5% significance.
*** 1% significance.

without controls for structural housing characteristics (model 1b in Table 5). The results are reassuring. While the explanatory power falls from 76% to 48%, the main point estimate changes only slightly. This suggests that the untreated areas present a reasonable control group for the treated houses. To provide extra support for this conclusion, we estimate the model on a sample containing houses on the affected streets only (model 1c in Table 5). This also yields a point estimate that is very close to the baseline estimate and is significant at 5%.

Furthermore, we check the robustness of our results by including only two years in the estimation, 1999 and 2006. For these years we have traffic density measurements; other years' traffic is imputed (see Section 2). Here again, the point estimate does not change significantly, although the sample size reduces eightfold and the coefficient loses significance (model 1d in Table 5).

Finally, model 1e in Table 5 reports the results of estimating Eq. (2) with anticipation effects for dwellings that experienced a decrease in traffic density. The point estimate of the anticipation coefficient β_2 has a correct negative sign, a low absolute value and is not significant. The point estimate of the price elasticity to density (−0.016) lies well within the confidence band of the baseline model. This supports the conclusion of no significant anticipation effects. This result is in line with the insights from Fig. 4 above.

5. Implications for the transportation policy appraisals

5.1. Comparison with conventional estimates

Revealed-preferences hedonic methods have been widely used to derive the economic value of traffic nuisance (see e.g. Navrud, 2004). Conventionally, the hedonic method implies a regression of prices based on the traffic nuisance and all other attributes of the dwelling including structural and neighbourhood characteristics, using a (pooled) cross-section of housing data. Recently, the economic profession has been expressing increasing concerns about the validity of cross-sectional studies for deriving valuation of environmental goods. Unobserved neighbourhood characteristics tend to be correlated with housing prices and the environmental good of interest, leading to a bias in cross-sectional estimates. This is especially true for the valuation of traffic nuisance as traffic density is a characteristic of a street as a whole; it is thus likely to be correlated with other characteristics of this street. Thus, in the case of traffic nuisance, the quasi-experimental approach we use in this paper should convey important advantages in comparison with a conventional approach.

We have tested this by running a conventional specification of Eq. (1) based on pooled cross-sectional data for the years 1998–2000.[16] In the estimation we include similar covariates to Van Ommeren et al. (2011). Appendix B reports the findings, while Table 6 (below) summarizes the main results. The first thing that stands out in comparison with the quasi-experimental fixed effects estimation, is the large difference in the coefficients of structural housing characteristics. In a pooled cross-section estimation, coefficients of housing attributes obviously account for some correlated unobserved location characteristics that in the quasi-experimental design are taken care of by the fixed effects. A larger coefficient of the size of the home (0.89 in a pooled OLS versus 0.55 in a quasi-experimental specification) suggests, for example, that larger houses are located in more attractive neighbourhoods. As traffic density is likely to be correlated to other (unobserved) neighbourhood characteristics, strong endogeneity of housing covariates suggests that traffic density estimates are likely to be biased too.

It is rather difficult to say in advance whether to expect a positive or negative omitted variables bias for traffic density. One can think of a number of undesirable unobservables connected with living on a busier (larger) street, such as more noise from people and activities, crime due to easier access or pollution. On the other hand, a larger street may also have some positively valued unobserved characteristics: a wider view out of the windows, larger distance to houses on the opposite side, presence of trees or better transport accessibility. The estimate from the pooled OLS is lower than the quasi-experimental estimate, with the difference being somewhat larger than two standard deviations. This could be interpreted as a sign that, in our dataset, streets with higher traffic density are ceteris paribus more attractive. However, this may also be an artefact following from the endogeneity of housing covariates and the correlation of traffic density with other neighbourhood characteristics. We can test this empirically by using the estimated coefficients from our quasi-experiment. The estimated postcode fixed effects f_j account for unobserved location characteristics of a postcode unit. The correlation of f_j with the log traffic density is positive and large (0.43 for the through streets with measured traffic density). Thus, in our dataset, locations with desirable unobservables have more traffic, and this creates

[16] The three years are pooled around the year 1999 for which we have measurements of traffic density. (An estimation for 1999 only yields similar results, but the sample is smaller and the coefficient on traffic density becomes non-significant).

Table 6
Quasi-experimental versus conventional estimates, selected coefficients.

	Quasi-experiment, postcode fixed effects 1998–2006	Pooled cross-sectional estimation 1998–2000
Selected structural housing attributes		
Living area in m^2 (ln)	0.549***	0.868***
	(0.020)	(0.023)
Number of rooms (ln)	0.100***	0.149***
	(0.014)	(0.021)
Traffic density (ln)	−0.019***	−0.005*
	(0.005)	(0.003)

* 10% significance.
** 5% significance.
*** 1% significance.

a downwards bias in the OLS estimates making it look like people are not inclined to pay to avoid traffic.[17]

5.2. Comparison with existing European guidelines

The valuation of traffic nuisance derived in this paper comprises the total of various negative traffic externalities that people perceive as nuisance. These include noise annoyance, pollution, accident risks and parking problems. In applied *ex ante* project appraisals, different effects of traffic are usually estimated separately and added up afterwards. This approach may convey inaccuracies, especially when the monetary values attached to different externalities have been estimated using divergent methodologies. Our results allow for a more consistent calculation of the total impact of different partial effects of urban traffic nuisance.

There is a large literature on the valuation of traffic noise annoyance, as noise is often perceived to be the major source of traffic nuisance. The European Commission (2003) recommends using a traffic noise annoyance valuation of 25 euros per decibel per household per year. Below we compare this valuation parameter to the nuisance valuation derived from the results of our study.

For medium density streets we have found a price elasticity to density of −0.02. This implies that a 50% reduction in traffic nuisance corresponds to a 1.4% increase in housing prices.[18] For our data, this effect equals some 2800 euro per household.[19] This is the capitalized benefit from halving the traffic nuisance. To derive the yearly benefit, one has to assume a discount rate and take into account different fiscal regulations applicable to the home owners, such as the deductibility of mortgage interest and property tax payments from taxable income. A recent Dutch study (Van Dijk and Romijn, 2010) calculates the gross discount rate for home owners to be 5.5% and the net discount rate to be 4.2%. Using this insight, the annual benefit from halving traffic nuisance is 116 euro per household, in prices in the year 2003.

Halving traffic density corresponds to some 3 decibels less noise annoyance.[20] According to the above mentioned recommendations of the European Commission the valuation of 3 decibels of noise is $25 \times 3 = 75$ euro. Our valuation is some 1.5 times higher, supporting the notion that traffic nuisance comprises more factors than noise annoyance only.

6. Conclusions

Our study provides new and convincing evidence on the valuation of urban traffic nuisance. We implement a strong test based on a quasi-experimental change in traffic density on local streets as a result of the construction of a new bypass in the west of the Netherlands. We use a unique dataset that not only contains information on which dwellings were affected by the traffic density change, but also shows to what traffic density levels these dwellings were exposed in the before- and after-periods. We derive monetary valuation for the total of various negative urban traffic externalities that people perceive as nuisance, including congestion, noise, local pollution and accident risks.

We find that on medium density streets the elasticity of prices to traffic density amounts to −0.02. On streets with high density the valuation of traffic nuisance may be up to 5 times as high as on the medium density streets. The impact of the traffic nuisance is furthermore not only present for the houses adjacent to the main street, but also for those located further away, up to a distance of 20 m from the street. We show that our estimates of the impact of traffic density changes are robust across space and across different house price segments.

As car drivers themselves do not internalize the costs they impose on others, local authorities in many urban agglomerations take various policy measures to reduce urban traffic flows. These include congestion charges, (high) parking costs as well as investment in tunnels and bypasses. Many such measures involve a substantial cost to car owners and taxpayers, so successful interventions require a careful appraisal of the costs and benefits of the resulting traffic reduction. Our paper offers a better insight into the benefits of these policies.

Our research suggests that the combination of quasi-experiments with sales transactions panel data offers a powerful tool for deriving the valuation of environmental and other non-market goods. Traditionally, a multiple regression is used to reduce the omitted-variable bias in such valuations. By comparing results from a conventional multiple regression analysis with quasi-experimental results, we argue that simple cross-sectional studies are likely to produce biased results and should be treated with caution.

One can compute the benefits of traffic nuisance reduction brought about by the bypass studied in this paper. Following a conservative approach that only accounts for the effects on houses adjacent to the affected streets, and takes the medium density elasticity of −0.02, the total local benefits for the houses in our dataset amount to 1.4 million euros in Leidschendam-Voorburg and 2.6 million euros in The Hague Mariahoeve. Assuming that the houses in the dataset are a random sample from the housing stock, the total benefit of reduced nuisance equals some 18 million euros.[21] The length of the bypass is 3.5 km, so the benefits are 5 million euros per kilometre bypass. To gain an insight into the relative importance of these benefits, we compare them with the reference construction costs of a two-lane bypass.[22] De Vos (2004) reports that a simple two-lane highway on ground level costs some 10 million euros per kilometre. Where tunnels and bridges are involved, as was the case with the bypass near The Hague, the cost easily quadruples to more than 40 million euros per kilometre (V&W, 2003). This simple calculation suggests that, for a bypass to be cost-efficient it must generate other benefits in addition to the reduction of urban traffic nuisance. For the bypass in our study an important benefit is the improved accessibility of The Hague.

[17] We can also correlate log density to the housing quantity index (predicted log price from Eq. (1) minus the estimated valuation of the traffic density). The housing quality index Q includes thus all (observed and unobserved) housing and locational characteristics except traffic nuisance and the unexplained residual. The index is calculated as: $Q_{ijt} = \hat{p}_{ijt} - \hat{\beta} \ln D_{jt}$, where $\hat{p}_{ijt}$ is the predicted value of log price for house i in postcode j in year t, D_{jt} is traffic density house i faces in year t and $\hat{\beta}$ is the estimated coefficient by traffic density. The correlation of Q_{ijt} with lnD_{jt} is 0.29 for through streets in our sample.

[18] $\Delta \ln P = -0.02 \times (\ln(1/2D) - \ln(D)) = -0.02 \times \ln(1/2) = 0.014$.

[19] The effect is valued at the average housing price of 197.000 euros.

[20] See, e.g., wikipedia.org/wiki/Decibel.

[21] The size of the housing stock as reported by Statistics Netherlands amounts to 34.2 thousand in Leidschendam-Voorburg and 8.4 thousand in The Hague Mariahoeve; the number of dwellings in our dataset for Leidschendam-Voorburg and The Hague Mariahoeve is 7.6 thousand and 1.8 thousand respectively.

[22] The bypass we studied was constructed by a private consortium, so its exact construction costs are unknown.

Appendix A. Descriptive statistics

Table A
Descriptive statistics: means, standard deviations are in parentheses.

	Affected dwellings, fall in traffic density		Affected dwellings, rise in traffic density		Not-affected dwellings	
Variable	Before	After	Before	After	Before	After
Price (× 1000 euro)	159,294 (119,617)	200,278 (143,149)	188,537 (118,903)	223,948 (149,633)	167,540 (117,514)	204,205 (145,937)
Traffic density (×1000 cars per day)	8.3 (8.6)	4.2 (4.9)	5.6 (3.6)	6.9 (3.3)	n/a	n/a
Structural attributes						
Living area in m^2	103 (49)	103 (46)	123 (49)	123 (50)	104 (42)	103 (42)
Number of rooms	4.1 (1.6)	4.1 (1.5)	4.8 (1.6)	4.7 (1.6)	4.2 (1.3)	4.1 (1.4)
Dummy apartment	0.78 (.41)	0.79 (.41)	0.77 (.42)	0.77 (.42)	.74 (.44)	0.74 (.44)
Dummy detached house	0.02 (.15)	0.02 (.14)	0.02 (.14)	0.01 (.08)	0.01 (.10)	0.01 (.11)
Dummy semindetached house	0.03 (.18)	0.03 (.17)	0.02 (.14)	0.03 (.16)	0.02 (.15)	0.02 (.15)
Dummy corner house	0.06 (.24)	0.06 (.23)	0.05 (.23)	0.05 (.23)	0.08 (.27)	0.07 (.25)
Dummy row house	0.10 (.30)	0.11 (.31)	0.14 (.34)	0.14 (.35)	0.15 (.36)	0.16 (.36)
Dummy year of construction <1905	0.01 (.09)	0.01 (.09)	0.01 (.08)	0.01 (.12)	0.00 (.06)	0.01 (.08)
Dummy year of construction 1906–1944	0.33 (.47)	0.33 (.47)	0.69 (.46)	0.66 (.48)	0.31 (.46)	0.28 (.45)
Dummy year of construction 1945–1970	0.50 (.50)	0.61 (.49)	0.23 (.42)	0.28 (.45)	0.42 (.49)	0.48 (.50)
Dummy year of construction 1971–1989	0.15 (.35)	0.04 (.20)	0.05 (.21)	0.02 (.14)	0.21 (.41)	0.18 (.38)
Dummy built-in garage	0.10 (.30)	0.10 (.30)	0.12 (.33)	0.07 (.26)	0.13 (.34)	0.10 (.30)
Dummy hot water heating	0.91 (.28)	0.93 (.26)	0.86 (.35)	0.93 (.25)	0.89 (.31)	0.93 (.26)
Dummy ground lease	0.39 (.49)	0.35 (.48)	0 (0)	0 (0)	0.13 (.34)	0.12 (.32)
Dummy nice view (water, open space)	0.39 (.49)	0.42 (.49)	0.32 (.47)	0.38 (.49)	0.41 (.49)	0.47 (.50)
Dummy Located in Leidschendam	0.45 (.50)	0.44 (.50)	0.91 (.29)	0.91 (.29)	0.57 (.50)	0.57 (.50)
Dummy Located in Voorburg	0.18 (.38)	0.20 (.40)	0.09 (.29)	0.09 (.29)	0.23 (.42)	0.23 (.42)
Dummy Located in Mariahoeve	0.37 (.48)	0.36 (.48)	0 (0)	0 (0)	0.20 (.40)	0.20 (.40)
Dummy near bypass in tunnel	0.07 (.26)	0.09 (.29)	0 (0)	0 (0)	0.05 (.21)	0.04 (.20)
Dummy near bypass on ground level	0.23 (.42)	0.22 (.42)	0 (0)	0 (0)	0.04 (.20)	0.04 (.19)
Land use and socio-economic characteristics neighbourhood						
Number of monuments within 1 km	8.1 (11.0)	7.5 (9.7)	6.5 (11.0)	5.6 (8.7)	8.8 (12.0)	8.5 (12.0)
% non-western immigrants	13 (6.6)	16 (7.0)	12 (3.7)	14 (3.5)	13 (6.8)	15 (8.6)
Average household income ×1000	14 (1.9)	15 (1.3)	14 (1.8)	15 (.8)	14 (2.0)	15 (1.8)
Population density ×1000	7.3 (2.3)	7.3 (2.2)	9.0 (1.3)	9.1 (1.1)	7.5 (2.3)	7.4 (2.3)
Distance to downtown The Hague	4.0 (1.2)	4.0 (1.2)	3.0 (1.2)	3.0 (1.2)	3.9 (1.4)	4.0 (1.4)
Distance to closest rail station km	0.8 (0.5)	0.8 (0.4)	0.9 (0.5)	0.9 (0.4)	0.9 (0.5)	1.0 (0.6)
Distance to closest on-ramp km	1.0 (0.4)	1.0 (0.4)	1.0 (0.5)	1.0 (0.5)	1.2 (0.6)	1.3 (0.6)
Number of observations	799	415	281	150	4409	2686

Appendix B. Coefficient estimates baseline model and pooled OLS regression 1998–2000

Table B
Coefficient estimates baseline model and pooled OLS regression 1998-2000.

	Quasi-experiment		Pooled OLS	
	Coefficient	St. dev.	Coefficient	St. dev.
Traffic density (ln)	−0.019***	0.005	−0.005*	0.03
Structural housing attributes				
Living area in m^2 (ln)	0.549***	0.020	0.868***	0.023
Number of rooms (ln)	0.100***	0.014	0.149***	0.021
Area in m^2 (ln)	0.041***	0.007	0.017**	0.008
Dummy apartment	Reference		Reference	
Dummy detached house	0.368***	0.047	0.354***	0.059
Dummy semidetached house	0.160***	0.039	0.243***	0.050
Dummy corner house	0.127***	0.035	0.128***	0.041
Dummy row house	0.083**	0.034	0.068**	0.039
Dummy year of construction <1905	−0.114*	0.057	−0.234***	0.062
Dummy year of construction 1906–1944	−0.072**	0.038	−0.261***	0.022
Dummy year of construction 1945–1970	−0.098***	0.038	−0.226***	0.021
Dummy year of construction 1971–1989	−0.106***	0.037	−0.244***	0.022
Dummy year of construction 1990–2006	Reference		Reference	
Dummy built-in garage	0.070***	0.009	0.139***	0.014
Dummy hot water heating	0.083***	0.006	0.096***	0.013
Dummy ground lease	−0.022*	0.012	−0.048***	0.018

(continued on next page)

Table B (*continued*)

	Quasi-experiment		Pooled OLS	
	Coefficient	St. dev.	Coefficient	St. dev.
Traffic density (ln)	−0.019***	0.005	−0.005*	0.03
Dummy nice view (water, open space)	0.007*	0.004	0.010	0.009
Location attributes				
Dummy located in Voorburg 2004–2006	Reference			
Dummy located in Leidschendam 2004–2006	0.039***	0.007		
Dummy located in The Hague Mariahoeve 2004–2006	−0.007	0.010		
Neighbourhood amenity bypass in tunnel 2004–2006	0.018	0.013		
Neighbourhood amenity bypass on ground level 2004–2006	−0.008	0.042		
Number of monuments within 1 km			0.002***	0.001
% non-western immigrants in the neighbourhood			−0.007***	0.001
Located in Voorburg			Reference	
Located in Leidschendam			0.100***	0.025
Located in The Hague Mariahoeve			0.102***	0.024
Distance to centre of the Hague (km)			0.041***	0.008
Distance to closest (light) rail station			0.020*	0.011
Distance to closest on-ramp			−0.006	0.013

* 10% significance.
** 5% significance.
*** 1% significance.

Appendix C. Robustness checks baseline model

In this appendix we test whether the estimates of model (1) stay robust when we allow for different valuations of traffic nuisance in different housing market segments. We investigate regional and price segmentation. To control for possible income differences in nuisance valuation,[23] we estimate separate regressions for houses in a higher and lower price segment.[24] To control for possible regional differences, we estimate separate regressions for each of the three towns in our dataset. The estimated traffic nuisance effect is robust across space and income.

Table C
Robustness analysis for market segments, standard errors in parentheses.

Traffic density (ln)	(1) Voorburg	(2) Leidschendam	(3) The Hague Mariahoeve	(4) Low Income	(5) High Income	(6) One-family homes	(7) Apartments
Adjacent to street	−0.025*	−0.024	−0.019***	−0.018***	−0.019*	−0.020	−0.019***
	(0.014)	(0.020)	(0.006)	(0.006)	(0.010)	(0.023)	(0.006)
PC6 fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Neighbourhood-specific time trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Housing attributes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2 within	0.77	0.76	0.80	0.77	0.77	0.77	0.74

* 10% significance.
** 5% significance.
*** 1% significance.

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[23] Day et al. (2007) suggest that people with higher income have a higher valuation of the amenity of quiet.
[24] The higher price segment is defined as all the houses with prices above the average price, the lower price segment is defined in a similar way.

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